The Control Is the Instrument

Calibrating Perturbation Tests Used as Evidence That EEG Decoders Read Neural Signal

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A standard argument in brain–computer-interface research runs: we removed the μ band, accuracy dropped, therefore the decoder uses sensorimotor rhythm. The argument is valid only if the perturbation has known sensitivity and, critically, a known false-alarm rate for the decoder being tested. Almost no published perturbation control has been characterised that way. We treat controls as measuring instruments and calibrate them against operational ground truth. Using a simulator in which the signal is planted by construction, we measure each control's catch rate and false-alarm rate across a grid of conditions and four decoder families. The reliability of a control turns out to depend on the decoder, not just on the control. Averaged over probes, EEGNet's false-alarm rate is 1.3% while band-power's is 68.8%: on the same tested simulator grid, the identical evidence has sharply different reliability across decoders. These full-grid rates are descriptive because each condition has one generated dataset. In a separate 30-dataset replication of the minimal comparison, conclusions are unchanged for firing thresholds 0.05–0.20. A random forest reading exactly the same features as a band-power linear model cuts spatial false alarms roughly in half while remaining equally fooled by band perturbations, implicating both representation and model class without identifying their interaction. Applying decoder-specific calibration to real PhysioNet drops changes their evidential weight. Finally, we grade the controls on real brains using eyes-open versus eyes-closed, a task whose occipital-alpha ground truth is fixed by physiology: band-power false alarms on three of four controls whose targets are absent and CSP on two, corroborating the specificity warning without establishing motor-imagery ground truth.

CCS Concepts: • **Applied computing: Life and medical sciences** • **Computing methodologies: Machine learning** • **Computing methodologies: Modeling and simulation**

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Additional Key Words and Phrases: brain–computer interfaces, EEG, explainability, perturbation tests, simulation, validation

1 INTRODUCTION

Decoders that read motor imagery from scalp EEG are routinely defended with perturbation evidence. The analyst removes a frequency band, scrambles or silences a set of electrodes, and reports that accuracy fell; the fall is then offered as proof that the decoder depends on genuine neural activity in that band or region. The logic is appealing because it appears causal—an intervention was performed and an effect was observed.

The logic is incomplete. A perturbation test is a diagnostic instrument, and an instrument that has never been characterised has no known error rate. To interpret "the control fired" one needs two numbers: the probability that it fires when the targeted dependence is genuinely present (sensitivity), and the probability that it fires when the targeted dependence is genuinely absent (false-alarm rate). Without the second number, a positive result is uninterpretable, because a control that fires under any perturbation whatsoever carries no information about which perturbation was applied.

Measuring those two numbers requires knowing the truth, and on real brains the truth is exactly what is unknown. This is the impasse we resolve by simulation.

Our concern parallels a line of work in machine-learning interpretability showing that attribution methods can pass visual inspection while failing basic sanity checks [1, 7], that occlusion-style evaluations need careful controls [6], and that model weights in neuroimaging admit no direct interpretation as signal sources [5, 13]. Perturbation controls in BCI have not received the same scrutiny.

1.1 Contributions

(1) An instrument-calibration framing of perturbation controls, in which a control is characterised by a measured (sensitivity, false-alarm) pair rather than assumed valid because it is intuitive.
(2) A controllable MI-EEG simulator with operational ground truth, and measured operating characteristics for band and spatial controls across four decoder families.
(3) The central empirical finding that control reliability is *decoder-conditional*, with large descriptive differences on identical data, plus controlled contrasts that vary feature scope and model class.
(4) A descriptive audit showing that the same PhysioNet accuracy drops receive different evidential weight after decoder-specific calibration.
(5) A real-brain validation on eyes-open/eyes-closed, a task whose ground truth is fixed by physiology, corroborating the simulator's specificity warning outside simulation while leaving motor-imagery ground truth unresolved.

2 A SIMULATOR WITH OPERATIONAL GROUND TRUTH

2.1 Generative model

Each trial is a matrix $X \in \mathbb{R}^{C\times T}$ with $C = 19$ electrodes at $F_s = 128$ Hz over $T = 256$ samples (2 s). The background is per-channel pink noise, matching the $1/f$ character of real EEG [14]. For real-FFT bins $k = 0, \dots, T/2$, let $f_k = kF_s/T$, draw $A_{ck}, B_{ck} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, and define

$$Z_{ck} = (A_{ck} + iB_{ck})q_k, \qquad q_0 = 1, \quad q_k = f_k^{-1/2}\ (k > 0), \qquad n_c = \operatorname{irfft}(Z_c), \quad n_c \leftarrow \frac{n_c}{\operatorname{std}(n_c) + 10^{-9}}. \tag{1}$$

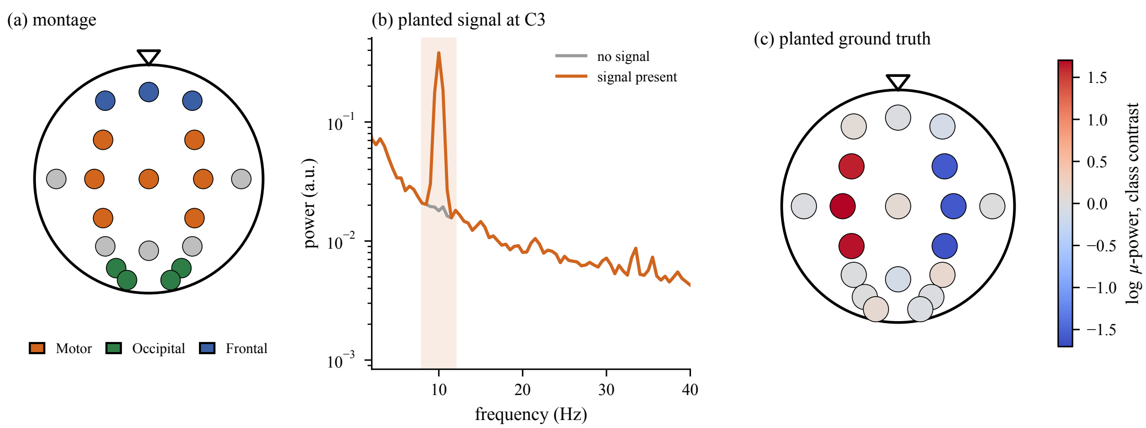

Fig. 1. The simulator. (a) The 19-electrode montage with region labels. (b) Welch spectra at C3 with and without the planted signal; the μ-band bump is present only when $s > 0$. (c) The planted ground truth: the class contrast in μ power is confined to motor channels by construction.

The inverse real FFT supplies the conjugate negative-frequency bins; its DC and Nyquist bins are real (equivalently, their imaginary parts are discarded). Thus the DC term is finite rather than divided by zero, and each output channel is normalised to unit sample standard deviation without an additional mean-subtraction step. The planted signal is a windowed 10 Hz μ burst added only to the motor channels contralateral to the imagined hand,

$$X_{c,t} = n_{c,t} + s\,\mathbb{1}[c \in \mathcal{M}(y)]\;\; w(t)\sin\!\big(2\pi f_\mu t/F_s\big)\,, \tag{2}$$

where w is a Hann window, $f_\mu = 10$ Hz, $\mathcal{M}(1)$ is the left-motor set and $\mathcal{M}(0)$ the right-motor set, and s is the signal-strength knob. Setting $s = 0$ produces data with *provably* no class-discriminative signal anywhere, which is the regime in which false alarms are defined. Equation (2) reproduces the lateralised event-related desynchronisation that motor-imagery decoders exploit [11].

Figure 1 shows the testbed: the montage, the spectral effect of the planted burst, and the resulting ground-truth topography, which sits on the motor channels and nowhere else.

2.2 Ground truth is operational, not biological

We are explicit that Equation (2) is not a cortex. Its value is *identifiability*: because we placed the signal, we know precisely which perturbations target something that is there and which target something that is not. That is exactly the knowledge real data cannot supply, and it is the minimum needed to measure a false-alarm rate at all. Section 5 addresses how far the resulting calibration transfers.

The construction is deliberately easier than biological EEG: background channels are independent, the carrier is fixed at 10 Hz, amplitude and onset are constant, and no volume conduction or artefact process is simulated. Spatial correlation could make a regional ablation less specific; frequency jitter could spread a true dependence across bands; and amplitude or onset variation could reduce sensitivity. Absolute rates should therefore be re-estimated under correlated noise, jitter, and temporal variability before transfer to another pipeline. The present simulator tests calibration logic, not biophysical realism.

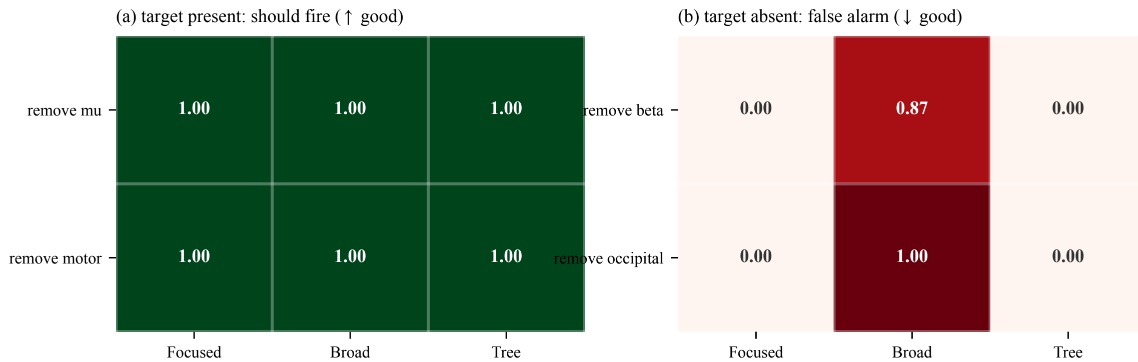

Fig. 2. Measured fire rates over 15 simulated datasets. (a) All three decoders detect the genuinely planted dependence. (b) The linear Broad decoder false-alarms on 87% of β removals and 100% of occipital removals, where no signal exists. Tree sees the identical 38 features and never false-alarms.

3 CONTROLS AND THEIR OPERATING CHARACTERISTICS

3.1 Definitions

A control is a map $\mathcal{P} : X \mapsto X'$ that ablates a hypothesised carrier. Band controls zero a spectral interval,

$$\mathcal{P}_{\text{band}}(X) = \mathcal{F}^{-1}\left\{\tilde{X}(f) \cdot \mathbb{1}[f \notin B]\right\}, \tag{3}$$

and spatial controls silence an electrode set, $\mathcal{P}_{\mathcal{R}}(X)_{c,t} = 0$ for $c \in \mathcal{R}$. A control *fires* when the induced accuracy drop exceeds a fixed threshold,

$$\Delta = a(X_{\text{test}}) - a(\mathcal{P}(X_{\text{test}})) > \tau, \qquad \tau = 0.10. \tag{4}$$

Over repeated datasets we estimate the two rates that define the instrument,

$$\text{sensitivity} = \Pr(\Delta > \tau \mid \text{target present}), \tag{5}$$

$$\text{false alarm} = \Pr(\Delta > \tau \mid \text{target absent}). \tag{6}$$

A control is trustworthy for a given decoder only when (5) is high *and* (6) is low *for that decoder*. The 0.10 cutoff was fixed before evaluation as a practically conspicuous ten-point accuracy loss; it is an operating choice, not a physiological constant. We therefore report its sensitivity below rather than treating it as uniquely correct.

3.2 A minimal dissociation

Before the full grid, one controlled comparison isolates the mechanism. We plant the signal in motor-μ and build three decoders: Focused, a logistic model on two features (mean μ power over left- and right-motor channels); Broad, a logistic model on 38 features (μ and β power at all 19 channels); and Tree, a random forest on the *identical* 38 features. Two controls target the planted signal (remove μ, remove motor) and two target nothing (remove β, remove occipital). Results over 15 datasets appear in Figure 2.

All three decoders pass the sensitivity test: removing μ or silencing motor channels degrades every one of them, as it should. The false-alarm panel dissociates them completely. Broad fires on 86.7% of β removals and 100% of occipital removals—perturbations that provably destroy no class information—while Focused and Tree never fire.

Table 1. Minimal 19-channel replication over 30 independent datasets. FA denotes false-alarm rate; sensitivity is 1.00 for all three decoders at every threshold.

τ	Broad FA	Focused FA	Tree FA
0.05	0.933	0.000	0.000
0.10	0.933	0.000	0.000
0.15	0.933	0.000	0.000
0.20	0.933	0.000	0.000

The explanation is mechanical and worth stating precisely. A linear model assigns nonzero weight to essentially every input, so zeroing *any* feature shifts its decision value and can push trials across the boundary; the drop reflects input perturbation, not dependence on the ablated carrier. A tree only splits on the features it uses, so zeroing an unused band or region leaves its decision path untouched. Since Broad and Tree consume identical inputs, the difference cannot be attributed to what the decoder can see. Same information, different model class, opposite reliability.

3.3 Threshold and data-generation sensitivity

We repeated the minimal experiment on 30 independently generated train/test datasets (seeds 1000–1029) and swept $\tau \in \{0.05, 0.10, 0.15, 0.20\}$. Table 1 shows that the qualitative result is threshold-invariant in this range. Each rate pools the two present-target or two absent-target controls. For Broad, a dataset-cluster bootstrap gives false-alarm CI $[0.867, 0.983]$; for the zero rates, the dataset-level Wilson upper bound is 0.114, and unit sensitivity has lower bound 0.886. These intervals quantify independent data-generation variability, unlike retraining the same dataset.

3.4 Measured operating characteristics at scale

The 19-channel experiment isolates a mechanism with small named region masks. The full study expands the same pink-noise and region-specific injection scheme to a standard 64-channel montage, mapping motor and occipital masks to their homologous electrodes, and sweeps 50 conditions (rhythm × region × strength × confound × two signal types). It uses one generated dataset per condition and five model retrainings on that dataset. Thus retraining variation measures optimisation or resampling variation, not data-generation uncertainty.

The four families span distinct representations: band-power/logistic regression is a hand-engineered linear spectral model; CSP/LDA learns supervised spatial filters followed by a linear rule [12]; the forest supplies nonlinear interactions over engineered features; and EEGNet learns compact temporal and spatial convolutions directly from trials [8]. A control is graded only after its decoder clears a decodability floor. Figure 3 reports the result.

Sensitivity is uninformative—every cell is 83–100%, so every control detects real dependence. The discrimination is entirely in the false-alarm panel. Averaged over probes, the false-alarm rates are 68.8% for band-power, 63.0% for the tree, 24.7% for CSP, and 1.3% for EEGNet. Because the full grid does not independently regenerate each condition, these are descriptive rates without data-generation confidence intervals. They show strong decoder dependence in the tested configuration but should not be exported as universal constants or a universal factor-of-50 contrast.

The tree's position is instructive and refines the toy conclusion. At scale the random forest does *not* inherit the toy Tree's immunity: it halves band-power's occipital false alarms (33% versus 67%) but matches or exceeds it on band controls (75% on remove μ, 83% on remove β). Selectivity in the splitting rule protects against spatial ablation, where irrelevant channels are simply never split on, but not against band ablation, which corrupts the very features the tree

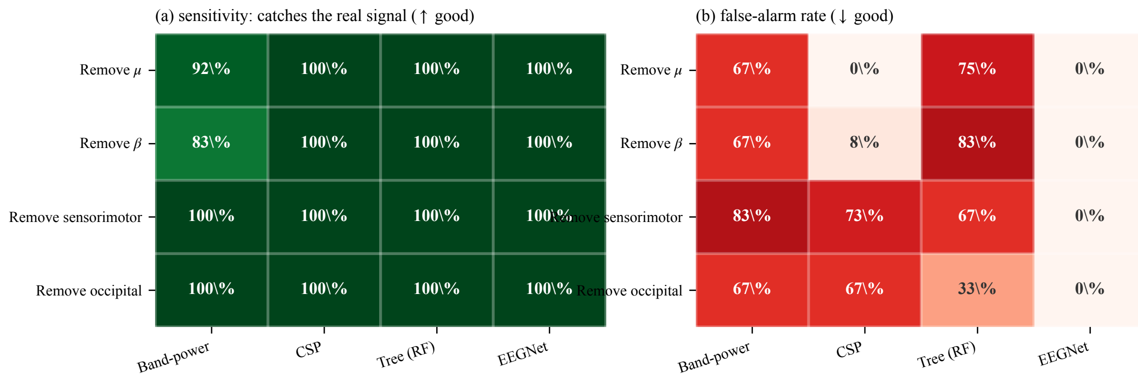

Fig. 3. Measured operating characteristics on the simulator grid. (a) Sensitivity is high everywhere: all four decoder families detect genuine dependence. (b) False-alarm rates differ by more than an order of magnitude. Band-power fires on 67–83% of perturbations whose targets are absent; EEGNet essentially never does.

does use. The failure therefore involves both feature representation and model class. The comparison is only a partial factorial design: Focused versus Broad varies feature scope within a linear model, and Broad versus Tree varies model class at fixed features, but a focused-feature tree is absent. A complete representation × model-class factorial is required to estimate their interaction.

4 AUDITING REAL DECODERS

We now apply the calibrated battery to decoders trained on PhysioNet EEGMMIDB [4, 15] with participant-held-out evaluation—every subject a decoder is tested on is one it never trained on. All three real decoders clear the decodability floor (Figure 4(b)); the BCI Competition IV-2a set [16] is excluded from the audit because nine participants render its test underpowered, which is a limitation rather than a result. The simulator cutoff in Eq. (4) defines event rates; the real audit retains continuous drops and attaches the corresponding decoder/control reliability rather than reapplying τ. Population inference must use participants, not trials, as the unit. Only aggregate held-out accuracies were retained, so participant-level confidence intervals cannot be reconstructed and the real-data comparison remains descriptive.

The verdicts are the paper's practical payoff. The four controls produce band-power drops of 0.03–0.08, and *none* is informative, because band-power's measured false-alarm rate on each is 67–83%. The same four controls applied to EEGNet yield drops of 0.01–0.09 with much lower measured false alarms: the pattern is consistent with dependence on μ and sensorimotor channels but not β or occipital channels—four configuration-specific indications, because EEGNet's false-alarm rate is near zero in the tested grid. CSP is intermediate, trustworthy on the band controls and suspect on both spatial ones.

An uncalibrated report of the same experiment would have listed eight accuracy drops of comparable magnitude and drawn eight conclusions. Four of them would have been unfounded.

5 VALIDATION ON REAL BRAINS

The obvious objection is that all of the above is simulated. Real data cannot in general settle it, because measuring a false-alarm rate requires knowing that a target is absent, and on real brains that is unknown—which is why the simulator exists.

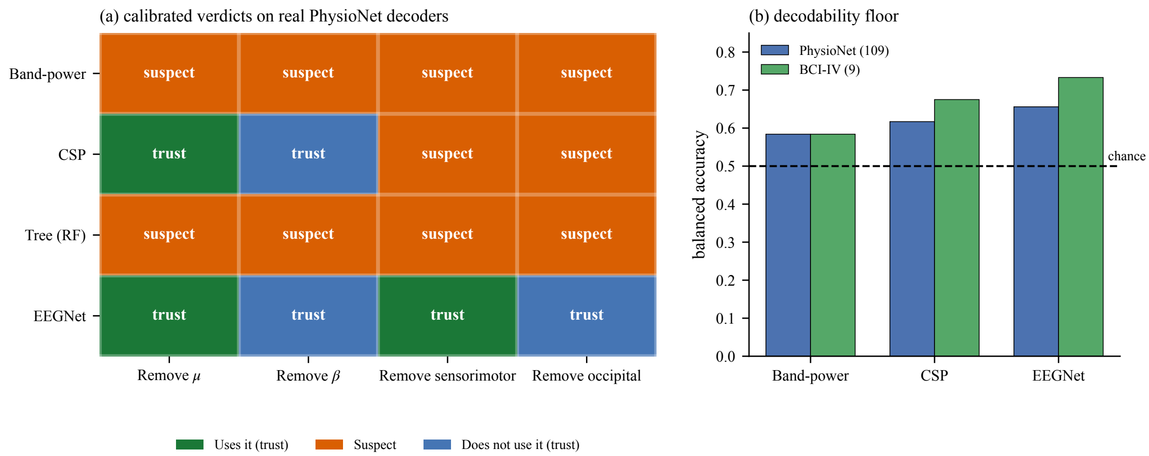

Fig. 4. (a) Calibrated verdicts on real PhysioNet decoders. Each cell attaches the measured reliability of Figure 3 to the observed accuracy drop, converting "the control fired" into a trust-weighted statement. Band-power and the tree earn no trustworthy verdict on any control; EEGNet earns four. (b) Held-out balanced accuracy: all decoders read motor imagery above chance.

There is, however, a way around the impasse: choose a real task whose ground truth physiology already fixes. Eyes-closed versus eyes-open is the strongest such case in electroencephalography. Closing the eyes reliably increases occipital alpha, an effect known since the field's founding [3] and characterised in detail since [2]. On the resting recordings of the same PhysioNet participants, we therefore know the discriminative signal is occipital and in the alpha/μ band. Controls aimed at that signal *should* fire; controls aimed at β, θ, sensorimotor, or frontal targets *should not*.

We decoded eyes-open versus eyes-closed on 34 subjects (band-power 73%, CSP 86% balanced accuracy—a real, strong signal) and graded the controls against that known truth. Figure 5 shows the outcome.

Band-power caught the genuine occipital-alpha dependence—and also fired on θ removal (+0.19), sensorimotor removal (+0.21), and frontal removal (+0.23), "detecting" three dependences that are not there. Its frontal false alarm is numerically as large as its true occipital detection (+0.23 each), so on this decoder the magnitude of a drop carries no information about whether the target was real. CSP false-alarmed on both spatial controls, with a frontal drop (+0.36) equal to its true occipital one.

This is a limited simulator-to-real bridge. It corroborates the specificity warning on a separate task whose physiology is independently known; it does not establish the same ground truth for motor imagery or validate the numerical simulator rates on real data. We note the natural confound—volume conduction spreads occipital alpha to distant electrodes [9]—and observe that it strengthens rather than weakens the conclusion: a control that fires because of field spread is exactly a control that cannot localise the signal it claims to localise.

6 DISCUSSION

6.1 What the study establishes

(1) Perturbation controls have measurable, *decoder-conditional* operating characteristics: descriptive full-grid false-alarm rates range from 1.3% to 68.8% on identical simulated data.

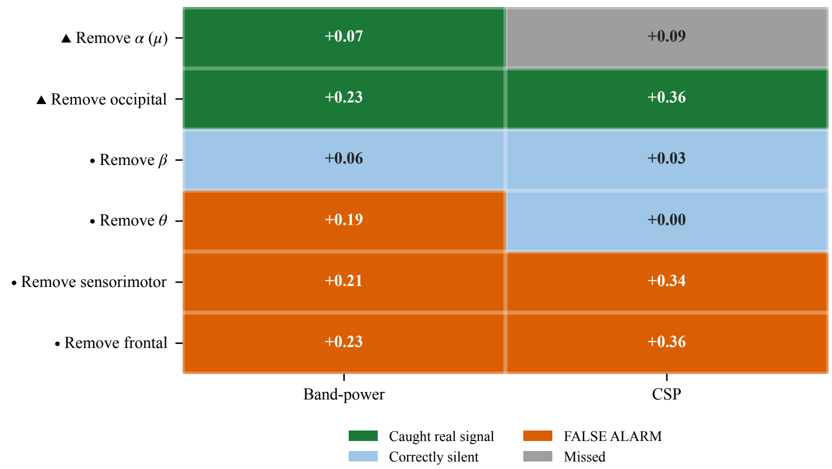

Fig. 5. Real-brain validation on eyes-open versus eyes-closed, 34 PhysioNet subjects, where physiology fixes the truth to occipital alpha. Triangles mark controls that should fire, dots those that should stay silent; cells give the accuracy drop. Band-power false-alarms on three of four absent targets, CSP on both spatial ones.

(2) Sensitivity is nearly uninformative in this setting; specificity does all the discriminating. Studies that report only that a control fired have reported the uninformative half.
(3) Controlled contrasts implicate both features and model class, but the incomplete factorial design does not identify their interaction.
(4) Calibration changes how continuous drops are weighted in the descriptive PhysioNet audit; participant-level uncertainty remains unmeasured.
(5) A known-physiology alpha task corroborates the specificity warning, not the motor-imagery ground truth.

6.2 What it does not establish

Simulated ground truth is operational, not biological, so how completely the measured reliabilities transfer to every recording pipeline remains open—this is the study's honest crux, not a footnote. The new independent-dataset analysis covers the minimal 19-channel comparison only. Each 64-channel grid condition still rests on one generated dataset, with five retrainings that measure model variability rather than data variability; its aggregate rates therefore lack data-generation intervals. Correlated spatial noise, frequency jitter, amplitude/onset variation, and a complete representation-by-model factorial remain to be tested. The real-brain confirmation covers one alpha task, not motor-imagery ground truth. Two older datasets, two-class motor imagery, and scalp EEG bound the scope. Trials evaluate a decoder, whereas participants support population claims, and no cortical-causality claim is made.

6.3 Recommendation

A perturbation result should not be reported as evidence unless the control's false-alarm rate has been measured for the decoder in question. Concretely, we recommend that papers using perturbation controls report, alongside each accuracy drop, the drop induced by at least one perturbation whose target is known to be absent for that decoder. That single addition converts an uninterpretable number into an interpretable one, and it is far cheaper than the calibration grid reported here.

7 CONCLUSION

Perturbation controls are instruments, and instruments require calibration. We measured the sensitivity and false-alarm rate of band and spatial controls against planted ground truth and found strong decoder dependence in the tested configurations. The independent 19-channel replication is stable across four thresholds, while the larger grid still needs independent dataset replication. Applying calibration to real-decoder drops changes their evidential weight, and a known-physiology task corroborates the specificity warning without establishing motor-imagery causality. The supported conclusion is methodological: perturbation evidence should be calibrated for its decoder and deployment pipeline.

REPRODUCIBILITY

The accompanying script regenerates the simulator panels and minimal calibration from Equations (1) and (2) using scikit-learn [10]. The original minimal study uses dataset seeds 0–14; each seed generates 60 training and 60 test trials per class, with test seed offset 999. Logistic models standardise features and use `max_iter=1000`; the forest uses 200 trees and seed 0; $\tau = 0.10$. The threshold analysis uses independent seeds 1000–1029, test offset 100,000, and $\tau \in \{0.05, 0.10, 0.15, 0.20\}$. CSV files retain aggregate 64-channel and real results, but not the per-retraining full-grid seeds or participant-level scores; this prevents reconstruction of those uncertainty estimates and is a reproducibility limitation.

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